Certified threshold decisions from error-bounded river water-quality records over one-way satellite IoT links

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Abstract

Satellite Internet of Things services reach remote rivers but sell only a few short, one-way messages per day, and the server never learns which messages were lost. This study asked what a server can decide about regulatory thresholds from the bytes that arrive. A node sends a self-contained error-bounded record within the accuracy that the U.S. Geological Survey requires of an excellent record, plus witnesses: quantised daily extrema and a conductance rate pair. A three-valued certificate rule then answers exceeded, not exceeded, or undetermined, and the answers are sound under stated assumptions. The method was developed on 38 rivers (2023) and confirmed once on 31 sealed rivers (2024) against nine frozen hypotheses. Eight held. No certified decision was wrong, and 30,285 message blocks passed a byte-level round trip without a mismatch. Targeted witnesses raised decisions on hard cases over a record alone by 0.30 to 0.32 at six messages per day with 30% loss. Decision flags beat witnesses on thresholds fixed at configuration time, but they decide from the flags alone. A hybrid that hosts witnesses in spare slots of the flag design cost nothing on configured thresholds and decided 0.08 to 0.10 more hard cases than the record alone, and 0.29 to 0.36 more than the flags, on thresholds that nobody configured. The ninth hypothesis, a bound on false certificates after a forged message that passes the 16-bit check, failed at twelve messages per day. The evidence is trace-driven; no satellite hardware was used.

Keywords: water-quality monitoring, satellite Internet of Things, error-bounded records, certified decisions, erasure coding, pre-registered evaluation

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1. Introduction

Continuous water-quality monitoring underpins the management of rivers that supply drinking water and receive pollution. In-situ sensors measure temperature, specific conductance, dissolved oxygen, and pH every few minutes, and the resulting records reveal diel cycles, storm responses, and short pulses that grab samples miss (Rode et al., 2016). Wireless sensor networks have made such monitoring affordable (Adu-Manu et al., 2017), and deployments in Ghana and Kosovo have shown that it works in practice, but they relied on cellular backhaul (Adu-Manu et al., 2020; Ahmedi and Ahmedi, 2022). The rivers under the greatest pressure are often the furthest from that infrastructure; in Ghana, for example, small-scale gold mining has degraded rivers in forested catchments where terrestrial coverage is sparse (Duncan, 2020). This motivating case is not evaluated here.

Satellite Internet of Things (IoT) services reach such sites with small, low-power terminals (Centenaro et al., 2021), but their capacity is sold in messages, not bandwidth. A typical module sends one-way messages with a payload of 23 bytes and receives no acknowledgement (Kinéis, 2023). A four-parameter sensor sampling every 10 minutes produces 576 readings a day; even packed at 13 bits each, without timestamps, into the 20 bytes that remain after a three-byte message header, they would fill at least 47 messages. Few such messages are affordable, some are lost, and the node is never told which. A monitoring agency therefore needs to know what the server can still conclude from what arrived. Many uses tolerate a delay of a day: compliance assessment, trend analysis, and model calibration work from daily or longer summaries, whereas alarms that require action within minutes are outside the scope of this study.

Agencies do not act on records; they act on decisions such as whether dissolved oxygen fell below 5 mg/L or whether a conductance pulse occurred. Delivering a whole record within an accuracy tolerance (Wagner et al., 2006) serves such decisions only partly. A record within tolerance cannot settle a reading that lies closer to a threshold than the tolerance allows, and when the quota is binding, bytes spent on the whole record are bytes not spent near the threshold. This study therefore took the decision as the object: the server states, for each day and criterion, one of three outcomes (exceeded, not exceeded, or undetermined), and the encoder spends its bytes so that the outcomes that matter are decided, with soundness that holds whatever the loss pattern, provided the messages that arrive are uncorrupted.

The study asked five research questions:

1. RQ1: Under which assumptions is a three-valued threshold decision from re-

ceived bytes sound, and what can no decoder decide?
2. RQ2: Do compact extremum witnesses, and targeted protection of the witnesses of hard cases, raise the share of decided hard cases over a record alone at a fixed byte quota?
3. RQ3: How does the witness design compare with decision flags (the answers to configured thresholds sent as bits), send-on-delta sampling, and a tightened tolerance, including on thresholds that were not known when the node was configured?
4. RQ4: What does it cost, in record recovery and in message capacity?
5. RQ5: How much does a message that is corrupted but passes its check endanger the soundness of the decisions?

The building blocks are established, and the study does not claim otherwise: minimum-size piecewise-linear approximation within a maximum error (Imai and Iri, 1988; O'Rourke, 1981); budgeted compression of sensor history (Deligiannakis et al., 2007); priority encoding transmission, which protects parts of a message unequally (Albanese et al., 1996; Boucheron and Salamatian, 2000); deterministic interval answers from compressed or uncertain data (Cheng et al., 2003; Brito et al., 2017; Lin et al., 2019a); possible-minimum and possible-maximum filtering of predicates (Taurone et al., 2024); and send-on-delta sampling (Diaz-Cacho et al., 2017). The contributions of this study are the following:

- A certificate language for a one-way lossy link: every piece of received information becomes an interval on a reading; threshold, extremum, conductance-change, and rolling-mean queries receive three-valued answers; the answers are sound when the intervals are valid (proved and checked exhaustively on small cases); and two limits are stated, namely that the rules are exact only without witnesses and that the tolerance makes near-threshold readings undecidable from a record alone.
- A message frame (version 2) with one CRC-16 per transmitted message, a witness section of 13-bit codes, and receiver rules (a sanity rule and corroboration), together with a byte-level codec and a mutation check of its tests.
- A comparison of designs on identical loss draws: record only, targeted witnesses (the 3b rule), an exact allocator, decision flags, a hybrid of flags and witnesses hosted in spare slots, send-on-delta, and a tightened tolerance, including negative results (the exact allocator did not beat the simple rule, and flags win on configured thresholds).
- A pre-registered confirmation on 31 sealed rivers in a year that development never saw, with nine hypotheses frozen before the data were opened. Eight held and one failed, and the failure is reported as it occurred.

Section 2 reviews related work. Section 3 describes the method. Section 4 describes the data and the pre-registered design. Section 5 reports the development and confirmation results, Section 6 discusses them and their limits, and Section 7 concludes.

2. Related work

Table 1 summarises the lines of work that touch this problem and states what each provides and what it does not provide for a one-way, byte-limited link. Most sources were read in part (abstract, introduction, method, and text searches); Albanese et al. was read for its definitions and its lower-bound theorem, and none of the sources was read in full.

2.1. High-frequency water-quality records and their quality

In-situ sensors record river chemistry every few minutes, which reveals storm responses, diel cycles, and short pollution pulses that monthly sampling misses (Rode et al., 2016). Such records are useful only if their quality is known: automated quality assurance and control has become part of streaming environmental data management (Campbell et al., 2013), and open software supports the visualisation, correction, and flagging of continuous records (Horsburgh et al., 2015). Agencies grade records by accuracy; the USGS rates a continuous record as excellent when its readings lie within narrow limits of the reference (Wagner et al., 2006). This study took those limits as the tolerance of the record and did not address sensor quality control, which remains upstream of the encoder.

2.2. Error-bounded reduction of sensor records

Sensor data reduction with guaranteed accuracy has two classical formulations. The first fixes a precision and minimises communication (Olston et al., 2003; Deligiannakis et al., 2004). The second fixes a budget and minimises error: buffered multi-signal history was compressed to a bandwidth budget with a maximum-error variant (Deligiannakis et al., 2007). For a single signal, the fewest vertices of a piecewise-linear approximation within a maximum error can be found efficiently (Imai and Iri, 1988; O'Rourke, 1981), also online (Soroush et al., 2008), and bounded-error approximation has been restricted to endpoints on a coarse grid (Lin et al., 2019b, 2024). Recent work preserves statistical features of the series, such as the autocorrelation function, within guaranteed bounds (Muñiz-Cuza et al., 2025), or supports analytics directly on a compressed base

(Sun et al., 2025). These works assume that what is sent arrives, and none certifies threshold decisions. This study used the minimum-number formulation as the encoder of the record and added witnesses and certificates around it.

2.3. Deterministic answers from compressed or uncertain data

Query processing over uncertain data answers with interval bounds and prunes values that cannot affect the answer (Cheng et al., 2003). Plato and its predecessor evaluate time-series expressions over segment-tree summaries of compressed series and return deterministic error intervals for aggregates (Brito et al., 2017; Lin et al., 2019a). Closest in mechanism, TREAT combines erasure coding with compression and uses the possible minimum and maximum of a value, computed from the bits that were retrieved, to filter predicates before download (Taurone et al., 2024). These works establish that interval answers and possible-extremum filters are classical. They address storage and retrieval, in which a missing fragment can be requested again; this study addresses a one-way link, in which the sender decides in advance what to protect and the receiver can never ask.

2.4. Priority encoding transmission and unequal protection

Priority encoding transmission (PET) lets the sender assign each part of a message a priority, the fraction of the encoding packets that suffices to recover that part. Albanese et al. proved that no system can have a total encoding length below the girth of its priorities and stated that systems reaching the girth exist for a priority function that closely approximates the requested one (Albanese et al., 1996); Boucheron and Salamatian related the approach to the capacity region of the broadcast erasure channel with degraded message sets (Boucheron and Salamatian, 2000). The layered group of witnesses in frame version 2 is an instance of this idea with two priority levels, namely the record, which needs K of Q messages, and a witness group, which needs k_w of Q. This study claims no coding result. PET decides how to protect parts that the sender has already defined; it does not decide which parts of a river record would settle a regulatory decision, and it has no notion of what the receiver can conclude.

2.5. Event-triggered telemetry and satellite practice

Send-on-delta sampling transmits a value when it leaves a deadband around the last sent value, with the threshold chosen for a target transmission rate (Diaz-Cacho et al., 2017); it is the standard event-triggered baseline, and this study evaluated it, alone and inside the record-first erasure code. Direct-to-satellite IoT has been studied mainly at the medium-access layer (Capez et al., 2022; Tondo et al.,

2021; Alvarez et al., 2022; Afhamisis and Palattella, 2022; Rathi and Borkotoky, 2025), with packets treated as interchangeable. The closest operational practice comes from animal telemetry: satellite relay data loggers abstract each ocean profile to a fixed number of broken-stick points and send the messages pseudo-randomly through the one-way Argos system (Boehme et al., 2009; Photopoulou et al., 2015). That practice fixes the number of points rather than the accuracy or the decision.

2.6. Goal-oriented communication, freshness, and reproducible environmental software

Goal-oriented and age-of-information research optimises the timeliness or the task performance of the latest value (Yates et al., 2021; Chen et al., 2021; Talli et al., 2026); it does not provide deterministic certificates of past decisions. Environmental modelling has long asked that models and software be developed and evaluated transparently (Jakeman et al., 2006), and computational hydrology has been criticised for results that cannot be reproduced (Hutton et al., 2016). Recent work in this journal defined levels of reproducibility (Essawy et al., 2020; Choi et al., 2021). This study therefore separated development from confirmation, froze its software and hypotheses before opening the confirmation data, and reports a failed hypothesis.

2.7. Position of this study

The only novelty statement that the searches support is limited in scope: a general web search limited to United States results, with the sources of Table 1 opened in part, found no earlier statement of a source-side encoder for a one-way lossy satellite link that, under a byte quota, selects an error-bounded record and feature-extremum witnesses and returns three-valued threshold decisions whose soundness is proved from the received bytes. The official bibliographic indexes were not searched, and the statement does not claim that no such work exists. Table 1 separates what was taken from earlier work from what was combined here.

3. Method

Fig. 1 summarises the method. A sensor node at a remote river site records four parameters. At the end of each day it builds a record of the day, adds witnesses, packs both into messages with erasure-coded redundancy, and sends a fixed number Q of messages through a one-way satellite link. The link delivers

Table 1: Related work and what it leaves open for a one-way, byte-limited link.

Line of work	Provides	Does not provide here
Error-bounded piecewise-linear approximation (Imai and Iri, 1988; O'Rourke, 1981; Lin et al., 2024)	Fewest vertices within a maximum error	Decisions, a byte quota, loss
Budget-first history compression (Deligiannakis et al., 2007)	Compression to a bandwidth budget with a maximum-error variant	One-way lossy links, decisions
Deterministic answers from summaries (Brito et al., 2017; Lin et al., 2019a)	Error intervals for aggregates over compressed series	Three-valued threshold answers, loss, selection of what to send
Interval queries on uncertain data (Cheng et al., 2003)	Pruning of intervals that cannot change an answer	Source-side encoding under a quota
Erasure coding with predicate filters (Taurone et al., 2024)	Possible-minimum and possible-maximum filters before download	One-way telemetry, a binding quota
Priority encoding transmission (Albanese et al., 1996; Boucheron and Salamatian, 2000)	Unequal protection with recovery from any fraction of packets	Which parts settle a decision; certificates
Send-on-delta sampling (Diaz-Cacho et al., 2017)	Transmission on leaving a deadband	Silent loss, certificates
Feature-preserving compression (Muñiz-Cuza et al., 2025; Sun et al., 2025)	Guarantees on statistical features; direct analytics	Threshold certificates, loss
Satellite tag practice (Boehme et al., 2009; Photopoulou et al., 2015)	Broken-stick points sent pseudo-randomly	Accuracy or decision targets, certificates

each message or loses it, and the node never learns which. The server decodes what arrived and answers, for each criterion, exceeded, not exceeded, or undetermined. This section defines the record (Section 3.1), the certificate language and what it can and cannot decide (Section 3.2), the witnesses (Section 3.3), the

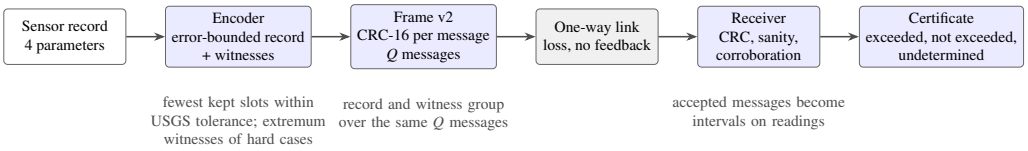


Figure 1: Workflow of the method. The node never learns which messages arrived; the receiver uses only accepted messages and the site profile.

message frame and receiver (Section 3.4), and the designs compared (Section 3.5).

3.1. Record, tolerance, and encoder

A channel-day is one parameter of one site on one UTC day, with readings $x(t)$, $t = 0, \ldots, n-1$, on the parameter's own sampling grid. The tolerance of a reading is $\delta(x) = \max(a, r|x|)$ with the accuracy that the USGS requires of an excellent continuous record (Wagner et al., 2006): 0.2 °C for temperature; 3% of the value for specific conductance; the greater of 0.3 mg/L and 5% of the value for dissolved oxygen; and 0.2 units for pH. The tolerance is relative to the sensor record; the accuracy of the sensor itself (calibration, fouling, and drift) is outside it.

The encoder selects the fewest kept slots whose piecewise-linear reconstruction keeps every reading within tolerance. This is the minimum-number problem of polygonal approximation (Imai and Iri, 1988) with per-reading tolerances, solved exactly by a forward dynamic program over slope windows (O'Rourke, 1981) in $O(n^2)$ time and $O(n)$ memory per channel-day. Three features make the bound exact and the record self-contained:

- Quantisation: every candidate error is computed with the dequantised endpoint values that the server will hold, such that the bound includes quantisation and needs no margin.
- Head and tail: the first kept value is held back to midnight and the last is held to the end of the day, such that no value of the previous day is needed.
- Censoring: a reading outside the quantiser range is always kept, carried as a flag code, excluded from every bound and decision, and reported as below or above range.

A kept reading is a 13-bit code. Linear grids are decimal (0.01 °C, 0.005 mg/L, and 0.002 pH units), and specific conductance uses a logarithmic grid with a step of 0.14% of the value; the worst quantisation error is 2.5% of the tolerance.

3.2. Certificate language

The language is independent of the codec: every source of information becomes a closed interval on a reading, and decisions are computed from intervals alone. The server holds, for every reading, an interval $[\mathrm{lo}(t), \mathrm{hi}(t)]$, a set of readings known to be censored, and optional global bounds. The knowledge is valid when every uncensored reading lies in its interval and every global bound holds. Truth statements refer to the sensor record that the loader produced, not to the river.

Sources of information and the intervals they provide:

- Base range: the quantiser range of the parameter, valid by the definition of censoring.
- Kept value with code c: the quantiser cell of c, which is far narrower than the tolerance (0 to 0.0025 units for temperature, dissolved oxygen, and pH, and 0.07% for conductance).
- Certified span: the tolerance interval around the reconstruction, valid when the day's bound flag is true and the bounding values of the span arrived. Lemma 1: for $r < 1$ and $x \geq 0$, the set $\{x : |x - \hat{x}| \leq \max(a, rx)\}$ is the interval $[\min(\hat{x} - a, \hat{x}/(1+r)),\ \max(\hat{x} + a, \hat{x}/(1-r))]$. The two sets whose union defines it share the point $\hat{x}$, such that the union is an interval with the extreme endpoints.
- Minimum witness: the cell of the kept extremum value, together with the assertion that every uncensored reading of the day is at least the lower end of that cell; the maximum witness is symmetric.

A reading with no information has the base range, such that it blocks a certain no.

Queries have three-valued answers: 1 (certainly yes), 0 (certainly no), or -1 (undetermined). Let $\mathrm{lo}_{\mathrm{eff}}(t)$ and $\mathrm{hi}_{\mathrm{eff}}(t)$ be the intervals intersected with the global bounds:

- Exceeds below θ (low dissolved oxygen, low pH): 1 if some $\mathrm{hi}_{\mathrm{eff}}(t) < \theta$; 0 if every $\mathrm{lo}_{\mathrm{eff}}(t) \geq \theta$; otherwise -1. The rule for exceeding above θ is symmetric.
- Day minimum and maximum: the minimum lies in $[\min \mathrm{lo}_{\mathrm{eff}}, \min \mathrm{hi}_{\mathrm{eff}}]$ and the maximum in $[\max \mathrm{lo}_{\mathrm{eff}}, \max \mathrm{hi}_{\mathrm{eff}}]$.
- Conductance change: some pair of readings one hour apart with a relative change above 10%. For readings with intervals $[L_a, U_a]$ and $[L_b, U_b]$, the ratio lies in $[L_b/U_a, U_b/L_a]$ and both ends are attained. A certain no holds when every pair is within 10% for all values of the intervals. A certain yes holds when no assignment of readings within their intervals keeps every pair within 10%; for one lag, the pairs form chains, and the set of readings reachable under "all pairs within 10%" is propagated along each chain until it is empty. The pairwise

rule "some single pair is certainly changed" is sound but incomplete (readings 100, an interval of 100 to 130, and 130 always contain a change, but no pair is certain); therefore, the chain rule is used.

- Seven-day mean of daily maxima above 20 °C (7DADM): the mean of the seven daily-maximum intervals lies in $[\text{mean of lower ends}, \text{mean of upper ends}]$. A day with no information contributes the base range, never zero and never omission; one missing day therefore blocks a certain no but not a certain yes that the other six days already force.

Proposition 1 (soundness): if the knowledge is valid, every answer 1 or 0 of every rule above equals the answer on the sensor record. Each rule evaluates the query on a set that contains the true record, and the extremum bounds follow because the minimum over readings is at least the minimum of the lower ends and at most the upper end of any reading. The ratio rule is the exact image of the box of intervals under the query, and the rolling mean is monotone in each daily maximum. Censored readings are outside the domain by definition. A censored reading that the server does not know to be censored receives the interval of a certified span; for the threshold queries this cannot produce a wrong answer, because the piecewise-linear reconstruction at such a position lies within the range of the neighbouring kept values, and the claim is not shown here for the conductance chain rule or for the daily maxima of the 7DADM. As a check, an exhaustive test over every subset of arrived values on small days, with and without each witness, found no interval that excluded the truth and no wrong answer.

Proposition 2 (exactness without witnesses, and its failure with them): for the threshold queries and the day extrema, when no witness has arrived, the rule returns an answer other than -1 exactly when the query has the same value on all records consistent with the knowledge. With a witness, the rules are sound but not exact. Counterexample: reading 1 in $[3, 10]$, reading 2 in $[4.5, 4.6]$, and a minimum-witness cell $[4.0, 4.01]$. Reading 2 cannot be the minimum; therefore, reading 1 lies in $[4.0, 4.01]$ and "some reading above 5" is certainly false, but the box rule returns -1. The decisions reported in this study are therefore a lower bound on what the witnesses could decide.

Proposition 3 (identifiability limit): if two records consistent with the received bytes have different answers to a query, no sound decoder can decide that query from those bytes. The tolerance makes this concrete. From a certified span alone, a threshold θ is decidable at a reading only when the reconstruction is farther from θ than the half-width of the interval (0.2 pH units, 0.2 °C, 3% of conductance, and the greater of 0.3 mg/L and 5% for dissolved oxygen). A reading closer to the threshold can be decided only if its value is carried, because the cell of a kept value

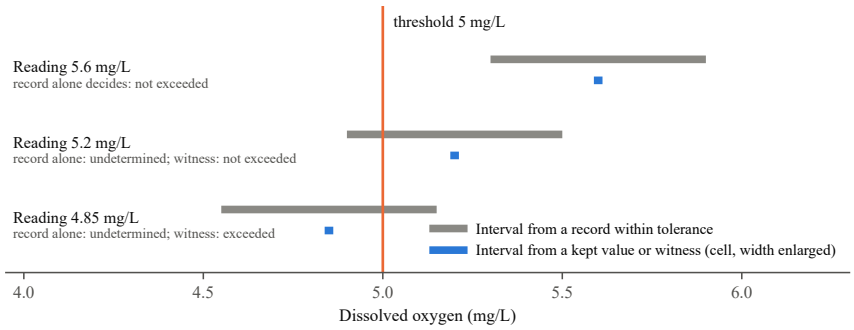


Figure 2: Constructed illustration (not data) of the identifiability limit for dissolved oxygen against a threshold of 5 mg/L. The tolerance interval of a record (grey) decides a reading far from the threshold but leaves readings near it undetermined; the cell of a kept value or witness (blue, enlarged for visibility) decides them.

is orders of magnitude narrower. This is the formal reason why a record within tolerance misses near-threshold events, and why bytes belong near the threshold and not everywhere. Fig. 2 illustrates the point with constructed values.

Soundness is conditional, not unconditional. It requires correct and corroborated messages, an honest encoder, a correct site profile, the quantiser range, and, where the rules use it, assumption R: readings are multiples of the site's reporting resolution. In this study the resolution was inferred from the analysed record itself, and the record contains readings that the loader bridged linearly over gaps of at most two readings. A wrong resolution is unsound: a reading of 7.03 with an assumed resolution of 0.1 yields a certain wrong decision for the threshold 7.01. In development, no resolution could be inferred for dissolved oxygen and pH at 15 of 38 sites, where assumption R was therefore off.

3.3. Witnesses and hard cases

A witness is the quantised extremum of one day and one parameter: the minimum of dissolved oxygen, the minimum and maximum of pH, the maximum of temperature (for the 7DADM), and the conductance rate pair, the two kept values that realise the day's largest hourly relative change. A witness costs 13 bits per code. It carries no slot, because no decision rule uses one.

Not every day needs its witnesses. A case is hard when the record alone is likely to leave the decision open:

- within 0.6 mg/L of the dissolved-oxygen threshold, within 0.4 pH units of a pH threshold, or in the event state;

Table 2: Message frame version 2 (23-byte payload).

Part	Bits	Content
Header (not coded)	24	version (2); block day modulo 64 (6); message index (7); K, the record data units (7); coded flag (1); witness-group flag (1)
Check	16	CRC-16 over the header and the rest of the body, in every message
Witness descriptor (if flagged)	8	k_w, the data units of the witness group (3); presence mask of the five witness objects (5)
Witness symbol (if flagged)	$8s_w$	one symbol of the witness group, with $s_w = \lceil \lceil 13 \cdot \text{codes}/k_w \rceil / 8 \rceil$ bytes
Record symbol	rest	one Reed–Solomon symbol of the record data units

- a conductance rate of at least 4% per hour;
- a 7DADM trailing mean within 3 °C of 20 °C, or with unknown history.

These bands were fixed on development data (the 7DADM band and the unknown-history rule after a first run fell short), and the confirmation used them unchanged.

3.4. Message frame and receiver

Each message carries the 23-byte user payload of the KIM1 module (Kinéis, 2023); the operator adds the device identifier and the time of reception (Kinéis, 2024). Table 2 provides the layout of frame version 2. A three-byte header (version, block day modulo 64, message index, K, and two flags) is not erasure-coded. The 20-byte body carries a CRC-16/CCITT-FALSE over the header and the rest of the body of every transmitted message, whether data or parity, an optional witness section, and one Reed–Solomon symbol of the record units over GF(2^8). The block day is unwrapped from the operator's reception time.

A record data unit carries up to three segments of kept values and a hosted-witness section: a flag, a 5-bit presence mask, and 13-bit codes in the spare room of the unit. Hosted witnesses cost no record capacity. Without witnesses a message holds five record values; one dissolved-oxygen witness in a layered group with $k_w = 1$ leaves four; four witness objects with $k_w = 2$ leave three. A computer-assisted enumeration over all 15 subsets of the four witness objects showed that, at equal protection, frame version 2 holds at least as many record values per message as the slot model used in the prototypes.

The receiver applies these rules:

1. Accept a message only if its CRC passes, and drop duplicates.
2. Decode the record from any K accepted messages and the witness group from any k_w accepted messages; with a check on every message there is no retry over combinations of parity messages.
3. Sanity: reject values outside the site profile or out of planned order, and censored witness codes.
4. Corroboration: when more messages arrive than are needed, re-encode the decoded units and compare with every other accepted message; any mismatch rejects the block, and nothing is certified from it. A witness descriptor that differs between messages, or two CRC-valid copies of one message that disagree, also reject the block.
5. Produce certificates by the rules above from accepted, corroborated content only.

Version 1 of the frame placed one CRC inside each erasure-coded unit; a parity message was then never checked alone, and every retry over another combination of parity messages was another chance for a corrupted message to pass. That design failed a pre-registered integrity hypothesis in an earlier confirmation, and version 2 is a new design, not a re-evaluation of the old one.

3.5. Designs compared

All designs send Q messages per day of five value slots each, use the same loss draws, and are evaluated by the same certificate rules.

- Record only: the record, with D_1 data units and $Q - D_1$ parity messages of a systematic Reed–Solomon code, such that any D_1 messages recover it.
- 3b rule: the generalised rule of this study. Witnesses of hard cases form a layered group in their own protected message columns; witnesses of easy cases are hosted in spare record slots. In the language of priority encoding transmission (Albanese et al., 1996), the record and the witness group are two priority levels over the same Q messages.
- Exact allocator (lambda 1): the allocation of witnesses to columns that maximises a stated objective under a prior over independent loss probabilities (0.1 to 0.5), solved exactly on a bounded subproblem.
- Decision flags: the answers to the four configured queries, sent as 8 bits in every message. The server decides a configured query from the flags whenever any message arrives and decides an unanticipated threshold only by monotone implication from the configured answer; the recovered record is not used for decisions. The prototypes modelled the flags as 4.75 record values per message,

against the integer 4 that frame version 2 allows, which favours the flags and also enlarges the spare room in which hybrid 1 hosts witnesses.

- Hybrid 1: decision flags for the configured queries, plus all witnesses hosted in spare record slots. It was adopted before the confirmation data were opened. The decision rule is the certificate decision when it is certain, otherwise the flag decision when any message arrived.
- Hybrid 2: decision flags plus the 3b rule. It was descriptive only.
- Random-day control: the same layering pattern as the 3b rule, applied to random days instead of hard days.
- Send-on-delta: a value is sent when it leaves a deadband around the last sent value, with hold reconstruction, sent once or inside the record-first code (Diaz-Cacho et al., 2017).
- Tolerance times 0.5: the record at half the tolerance, without witnesses.

4. Data and pre-registered design

4.1. Data and cohorts

All data were approved instantaneous values from the USGS National Water Information System (U.S. Geological Survey, 2016); dissolved oxygen was used in mg/L. Each parameter was used on its own sampling grid on each UTC day; a run of at most two missing readings with valid neighbours on both sides was bridged linearly, and any longer gap made that parameter-day ineligible. The analysis used four-parameter days. Two data sets served different purposes:

- Development: calendar year 2023 at 38 sites with four-parameter days (10,342 days) in New Jersey, Pennsylvania, Illinois, and Oregon. Every design choice, band, and rule was set on these data. Data from earlier rounds of the study (a 2024 re-test at the same sites and a sealed cohort of 36 sites in 2024 that had been used to confirm an earlier version of the method) are also development data for the present method and were not used as confirmation.
- Confirmation: calendar year 2024 at 31 sites that development never used, in four regions: the Pacific coast (9 sites), the Mid-Atlantic and Appalachia (9), the Ohio valley and Great Lakes (9), and the Gulf and lower Mississippi (4). Selection used metadata only: sites whose series of temperature, conductance, dissolved oxygen, and pH covered 2024 and that had at least 250 approved daily means per parameter (qualifiers read, values discarded) were taken in seeded random order within region, up to nine per region. Of 249 candidates, 31 were selected; only 4 of 16 Gulf candidates met the approval rule, and the shortfall was not repaired by adding regions. The data were downloaded, cleaned by

the rules of all earlier cohorts, and sealed with MD5 checksums before any analysis. The analysis population was the sites with at least 100 four-parameter days, which was all 31 sites (10,095 days, 144 to 365 per site).

4.2. Loss models, plans, and thresholds

No measured loss rate was available for the target service; therefore, loss was modelled. Independent loss removed each message with probability 0.3; a two-state Gilbert channel (Gilbert, 1960; Elliott, 1963) lost every message in its bad state with the same mean loss 0.3 and mean bursts of three messages. Plans of $Q = 6$ and $Q = 12$ messages per day were evaluated, with 20 loss draws per day and identical draws for all designs.

The configured thresholds were fixed before any design was compared: dissolved oxygen below 5 mg/L (U.S. Environmental Protection Agency, 1986a); pH below 6.5 or above 9.0 (U.S. Environmental Protection Agency, 1986b); a 7DADM above 20 °C (U.S. Environmental Protection Agency, 2003); and a conductance change above 10% in one hour (a criterion defined for this study). Two alternative threshold sets that the encoder does not know measure generality: alt1 (4.5 mg/L; pH 6.3 and 9.2; 21 °C; 20%) and alt2 (5.5 mg/L; pH 6.7 and 8.8; 19 °C; 5%). The analyst chose these sets, and the results that depend on them are reported with that caveat.

The comparison of designs used an abstract slot model, not the bytes of the frame: a message holds five value slots, a group of values over several columns is a systematic maximum-distance-separable code recovered from any k of the Q messages, and witnesses were applied with the slot of the extremum, whereas the frame carries the 13-bit code only. The byte-level tests (the round trip of H1 and the forgery study of H8) covered the record and a layered group of four witnesses; hosted witnesses, the conductance pair codes, the flags, and decoding from a subset of the messages were tested only in unit tests.

4.3. Hypotheses, freeze, and procedure

The study proceeded in two stages. In development, the method and the hypotheses with their numerical limits were written, and a separate AI session read the code and the plan adversarially (an in-project reading, not an outside review); several defects were corrected and a mutation check of the tests was run. In confirmation, the hypotheses (Table 3), the analysis plan, and the software were frozen: the repository was committed and tagged, a freeze record with checksums of every method file was written, and the confirmation script refused to run if any frozen file differed from the tag or if the seal did not verify. The tag was created on the

commit of the experiments on the instruction of the researchers, the freeze record was written when the run started, and it was committed after the run; the checksums in it were verified by the run script before any analysis. The sealed data were then opened once. The run took about 3.5 hours on one worker and was neither interrupted nor restarted.

Intervals are 95% percentile intervals from 1,000 site-clustered bootstrap resamples; differences between designs are pooled day-query means computed on the same resamples. Hard-case decisions pool dissolved oxygen, pH, conductance, and 7DADM windows, and a decision counts as correct only when it is certain and equals the truth on the sensor record. A hypothesis counts as supported only if its stated bound holds in every listed condition, and the limits were not changed after the data were opened.

Table 3: Pre-registered hypotheses and their outcome on the 31 sealed confirmation sites.

ID	Statement	Limit	Result (95% interval)	Outcome
H1	No wrong decision in any certificate design; no frame round-trip mismatch	0 wrong; 0 mismatches	wrong decisions max 0.000; round trip: 0 mismatches in 30285 blocks	Supported
H2	3b rule minus record only, hard cases at configured thresholds	lower bound at least 0.15	+0.318 (+0.234 to +0.417); +0.300 (+0.223 to +0.390)	Supported
H3	3b rule minus random-day layering, hard cases	lower bound at least 0.05	+0.141 (+0.105 to +0.174); +0.130 (+0.096 to +0.165)	Supported
H4	Record recovered, 3b rule minus record only	lower bound at least -0.12	-0.083 (-0.105 to -0.062); -0.071 (-0.091 to -0.053)	Supported
H5	Unanticipated thresholds, hard cases: 3b rule minus flags, and minus record only	both lower bounds at least 0.10	flags: +0.531 (+0.494 to +0.569); +0.441 (+0.411 to +0.472); record only: +0.260 (+0.191 to +0.344); +0.245 (+0.184 to +0.320)	Supported
H6	Neither send-on-delta nor tolerance x 0.5 dominates the 3b rule	no dominance	send-on-delta/independent: no; tolerance x 0.5/independent: no; send-on-delta/Gilbert: no; tolerance x 0.5/Gilbert: no	Supported

ID	Statement	Limit	Result (95% interval)	Outcome
H7	Known limit: flags minus 3b rule, configured thresholds, independent loss	lower bound above 0	+0.043 (+0.025 to +0.068)	Supported
H8	False-certificate day share given one CRC-valid forgery	upper bounds at most 0.06, 0.01, 0.15, 0.02	record, Q = 6: 0.0463 (upper 0.0584, limit 0.06); record, Q = 12: 0.0061 (upper 0.0134, limit 0.01); layered, Q = 6: 0.1050 (upper 0.1197, limit 0.15); layered, Q = 12: 0.0126 (upper 0.0215, limit 0.02)	Not supported
H9	Hybrid 1 minus flags: unanticipated thresholds; configured thresholds; record recovered	alt lower bound at least 0.20; others at least -0.01	alt: +0.355 (+0.297 to +0.418); +0.294 (+0.248 to +0.345); configured: +0.000 (+0.000 to +0.000); +0.000 (+0.000 to +0.000); record: +0.000 (+0.000 to +0.000); +0.000 (+0.000 to +0.000)	Supported

The statements of H2 to H9 refer to $Q = 6$ and to both loss models unless stated; H8 also covers $Q = 12$. H1 deserves one caution. Because the reporting resolution is inferred from the analysed record, H1 cannot fail through assumption R; it tests the code and the other assumptions, conditional on R being correct, and it is not evidence that R holds for deployed sensors.

5. Results

Section 5.1 summarises the development results that set the hypotheses; they are numerical and were produced on 2023 data. Sections 5.2 to 5.4 report the single confirmatory run on the 31 sealed sites. Pairs of values are independent loss and Gilbert loss, with loss probability 0.3 and $Q = 6$ unless stated.

5.1. Development results

On the 38 development sites (10,342 four-parameter days), the findings that shaped the pre-registration were the following:

- Soundness: no certificate design gave a wrong decision in any condition.
- Witnesses added decisions: the record alone decided 0.677 of hard cases at the configured thresholds (independent loss), and the 3b rule 0.968. A witness

placed inside the record code did almost nothing for the 7DADM, which needs seven informative days; placement in a protected group mattered.

- Targeting paid: against the same layering on random days, the 3b rule gained 0.10 to 0.15 on near 7DADM windows and 0.17 on hard conductance days, at a record-recovery cost of 0.001 to 0.005 in the single-query prototype and 0.008 (0.004 to 0.013) for the generalised rule.
- The exact allocator did not beat the simple rule on hard cases, and the pre-registered criterion for it was not met; the simple rule was within 7% (at $Q = 6$) and 1.6% (at $Q = 12$) of the optimum of the stated objective.
- Decision flags dominated on thresholds known at configuration time: flags decided 0.999 of hard cases against 0.968 for the 3b rule, and recovered the record more often. On two threshold sets that the encoder did not know, the 3b rule exceeded the flags by 0.42 to 0.57.
- Send-on-delta needed 1.79 times as many values as the piecewise-linear record, and a tightened tolerance did not improve hard-case decisions at this quota.
- Hybrid 1 matched the flags on configured thresholds and record recovery and gained 0.416 and 0.355 on the unanticipated thresholds; hybrid 2 lost record recovery. Hybrid 1 therefore met the adoption rule that was written in advance and was adopted before the freeze.

5.2. Confirmation: nine hypotheses

Table 3 lists every hypothesis with its limit, its result, and its outcome; eight were supported and H8 was not. The run completed without interruption: the freeze check passed, 31 of 31 checksums verified, and the log contains no error.

- H1, soundness: the maximum share of wrong certain decisions was 0.000 in every certificate design at $Q = 6$ and $Q = 12$ under both loss models, and the byte-level round trip of frame version 2 showed no mismatch in 30,285 blocks (10,095 days at three plans each, all messages delivered, covering the record and a layered group of four witnesses). The caveat of Section 4.3 applies.
- H2, benefit over the record alone: the 3b rule decided 0.318 (0.234 to 0.417) more hard cases at the configured thresholds under independent loss and 0.300 (0.223 to 0.390) under bursts, against a limit of 0.15.
- H3, value of targeting: against the same layering on random days, the gain was 0.141 (0.105 to 0.174) and 0.130 (0.096 to 0.165), against a limit of 0.05.
- H4, cost in record recovery: the 3b rule recovered the record less often than the record alone, by 0.083 (0.062 to 0.105) and 0.071 (0.053 to 0.091). The pre-registered limit was 0.12, and this was the hypothesis with the thinnest margin in development; it held, and the upper bound of the loss was 0.105 under

independent loss.

- H5, generality: on thresholds that the encoder did not know, the 3b rule exceeded the flags by 0.531 (0.494 to 0.569) and 0.441 (0.411 to 0.472), and the record alone by 0.260 (0.191 to 0.344) and 0.245 (0.184 to 0.320). The flags design does not use the recovered record; therefore, the first contrast is larger than the second; the contrast with the record alone is the fairer measure of the benefit of witnesses.
- H6, baselines: neither coded send-on-delta nor a tolerance multiplied by 0.5 dominated the 3b rule in any of the four combinations.
- H7, the known limit: the flags exceeded the 3b rule in hard-case decisions at the configured thresholds by 0.043 (0.025 to 0.068) under independent loss. This replicated the development finding and is a limit of the witness design, not a success.
- H9, the hybrid: hybrid 1 exceeded the flags on the unanticipated thresholds by 0.355 (0.297 to 0.418) and 0.294 (0.248 to 0.345), and matched them exactly on configured hard-case decisions and on record recovery (0.000 in all four contrasts). Those four contrasts are zero by construction, because hybrid 1 uses the same flags at the same capacity, and only the unanticipated-threshold contrast tests the hybrid. Over the record alone, hybrid 1 decided 0.083 (independent loss) and 0.097 (Gilbert loss) more hard cases at the unanticipated thresholds (descriptive point differences of the pooled levels, without intervals).

  Fig. 3 shows the contrasts with their intervals and limits.

5.3. Descriptive comparison of all designs

Table 5 and Fig. 4 show the pooled levels of every design at $Q = 6$; these results carry no verdict.

- The flags decided 0.999 of hard cases at configured thresholds under independent loss but only 0.275 on unanticipated thresholds; hybrid 1 kept the flag performance and raised the unanticipated level to 0.629 without reducing record recovery (0.877 for both).
- Hybrid 2 reached 0.806 on unanticipated thresholds, the same as the 3b rule, but recovered the record less often (0.779 against 0.877 for the flags).
- The exact allocator, relative to the 3b rule, changed record recovery by +0.028, hard configured decisions by -0.015, hard unanticipated decisions by -0.057, and all configured decisions by +0.049 under independent loss. As in development, it did not beat the simple rule on hard cases; therefore, no confirmatory claim is made for it.

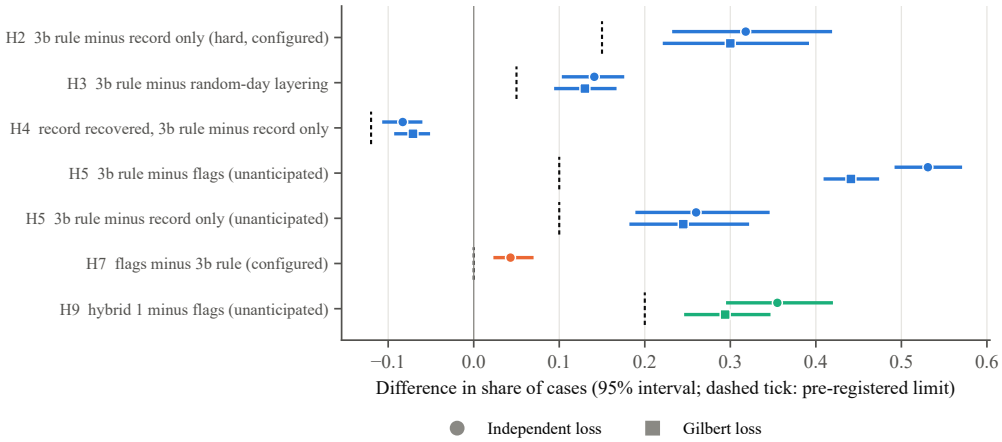


Figure 3: Confirmed contrasts at six messages per day with 95% site-clustered intervals; the dashed tick marks the pre-registered limit of each hypothesis. Circles are independent loss and squares are Gilbert loss. Blue marks contrasts of the 3b rule, orange the known limit (H7), and green the hybrid (H9).

- Send-on-delta and the tightened tolerance were below the record alone in record recovery and, with one exception, in hard-case decisions at this quota; the exception was the tolerance multiplied by 0.5 under Gilbert loss at the unanticipated thresholds (0.464 against 0.460).
- At $Q = 12$ (Table 6) the orderings were the same at higher levels: under independent loss the record alone recovered the record in 0.972 of days and the flags and hybrid 1 in 0.969, and the 3b rule decided 0.971 of hard cases at configured thresholds against 0.969 for the exact allocator and 1.000 for the flags and hybrid 1.

Table 7 splits the levels at $Q = 6$ by region, which was a pre-registered descriptive item. The regions have four to nine sites; therefore, the figures carry no interval and no verdict.

- The ordering on unanticipated thresholds held in every region and under both loss models: hybrid 1 and the 3b rule exceeded the flags, for example 0.578 and 0.849 against 0.368 in the Gulf region and 0.591 and 0.738 against 0.211 on the Pacific coast (independent loss).
- The cost in record recovery was not uniform. In the Gulf region, which has only four sites, the record alone recovered the record in 0.654 of days and the 3b rule in 0.488, a loss of 0.166 under independent loss; in the other regions the loss

ranged from 0.062 to 0.084. The pooled loss that H4 bounded (0.083) therefore hides a larger loss in one region. The reason was not investigated.

Table 4 splits the hard-case contrasts at $Q = 6$ by query kind, because the pooled hypotheses weight each kind by its number of hard cases. The figures are point differences of pooled levels, without intervals and without a verdict.

- Every contrast was positive for every kind under both loss models; therefore, no single kind drove the pooled results; for example, the 3b rule exceeded the record alone at the configured thresholds by 0.185 (dissolved oxygen) to 0.401 (conductance) under independent loss.
- The gain of hybrid 1 over the record alone at the unanticipated thresholds was uneven: 0.094, 0.135, and 0.085 for dissolved oxygen, pH, and the 7DADM under independent loss, but only 0.022 for conductance (0.031 under Gilbert loss). The pooled 0.083 therefore comes mainly from three of the four kinds.

Table 4: Hard-case contrasts at six messages per day by query kind (point differences of pooled levels; the numbers of hard cases at configured and unanticipated thresholds follow each kind): the 3b rule minus the record alone, and the 3b rule, hybrid 1 minus the flags and minus the record alone, at unanticipated thresholds unless stated.

Kind (hard cases)	3b − record, configured	3b − flags	3b − record	Hybrid 1 − flags	Hybrid 1 − reco
Independent loss					
Dissolved oxygen (1050; 2117)	+0.185	+0.544	+0.152	+0.486	+0.094
pH (2209; 5144)	+0.350	+0.594	+0.291	+0.438	+0.135
Conductance (1772; 4764)	+0.401	+0.368	+0.271	+0.119	+0.022
7DADM (1359; 2585)	+0.261	+0.697	+0.266	+0.516	+0.085
Gilbert loss					
Dissolved oxygen (1050; 2117)	+0.192	+0.501	+0.160	+0.444	+0.103
pH (2209; 5144)	+0.327	+0.535	+0.274	+0.398	+0.136
Conductance (1772; 4764)	+0.373	+0.330	+0.253	+0.108	+0.031
7DADM (1359; 2585)	+0.244	+0.410	+0.242	+0.304	+0.137

5.4. Integrity under a forged message: H8 failed

H8 bounded the share of days with a false certificate after one message that passes the CRC-16 and replaces an arrived message, given the full receiver (CRC, sanity, and corroboration). Two of its four bounds were exceeded; therefore, the hypothesis was not supported:

- Record only, $Q = 6$: 0.0463, with an upper bound of 0.0584 against a limit of 0.06. Within the limit.

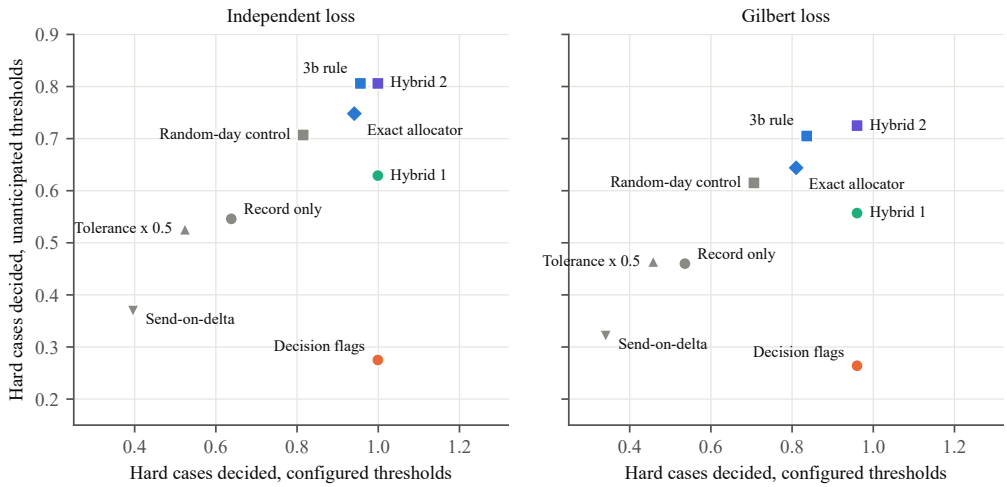


Figure 4: Trade-off between decisions at configured and unanticipated thresholds for every design at six messages per day (pooled hard cases). Flags decide almost every configured hard case but few unanticipated ones; hybrid 1 keeps the flag level on configured thresholds and raises the unanticipated level; the layered designs reach the highest unanticipated level.

- Record only, $Q = 12$: 0.0061, with an upper bound of 0.0134 against a limit of 0.01. Exceeded; the point estimate lay inside the limit and the upper bound did not.
- Layered witnesses with $k_w = 2$, $Q = 6$: 0.1050, with an upper bound of 0.1197 against a limit of 0.15. Within the limit.
- Layered witnesses with $k_w = 2$, $Q = 12$: 0.0126, with an upper bound of 0.0215 against a limit of 0.02. Exceeded, in the same way.

Fig. 5 compares the cohort with the development estimates and the limits.

Every cohort value was higher than the corresponding development value from six sites (0.027, 0.0004, 0.086, and 0.0029); the record-only value at $Q = 12$ was about 15 times higher. The false certificates came from three kinds of day, shown here for record only: days on which exactly the needed number of messages arrived (23% of the rate at $Q = 12$ and 58% at $Q = 6$), days on which fewer messages arrived than needed and only systematic units were used (40% and 24%), and days on which extra messages arrived but could not be compared, probably because the record exceeded the quota and had no parity (37% and 18%; this attribution rests on the receiver rules and the rejection pattern, not on a count of units per day, because uncoded blocks, which are the oversize days, are never corroborated). Development had almost no days of the second kind at $Q = 12$ (10 against 401

Table 5: Pooled levels of every design at six messages per day (record recovered, hard cases at configured thresholds, hard cases at unanticipated thresholds, and all days at configured thresholds).

Design	Record	Hard, configured	Hard, unanticipated	All, configured
Independent loss, $Q = 6$				
Record only	0.892	0.638	0.546	0.883
3b rule	0.809	0.956	0.806	0.923
Exact allocator	0.837	0.941	0.748	0.972
Tolerance x 0.5	0.698	0.524	0.526	0.754
Send-on-delta, coded	0.533	0.396	0.369	0.620
Send-on-delta, once	0.237	0.327	0.304	0.444
Decision flags	0.877	0.999	0.275	0.999
Random-day control	0.812	0.815	0.707	0.914
Hybrid 1	0.877	0.999	0.629	0.999
Hybrid 2	0.779	0.999	0.806	0.999
Gilbert loss, $Q = 6$				
Record only	0.817	0.536	0.460	0.783
3b rule	0.746	0.836	0.705	0.824
Exact allocator	0.759	0.810	0.644	0.869
Tolerance x 0.5	0.652	0.458	0.464	0.675
Send-on-delta, coded	0.520	0.341	0.321	0.567
Send-on-delta, once	0.355	0.302	0.284	0.453
Decision flags	0.804	0.960	0.264	0.960
Random-day control	0.747	0.706	0.615	0.816
Hybrid 1	0.804	0.960	0.557	0.962
Hybrid 2	0.723	0.960	0.725	0.963

in the cohort). The forgeries are random-content forgeries conditional on a CRC pass, and they changed body bytes only. The unconditional rate of false certificates depends on the real error rate of the channel, which is unknown, and was not tested in the confirmation.

6. Discussion

6.1. What the evidence supports

The confirmation supports a specific and bounded claim. On 31 rivers that development never used, the certificate rules were never wrong (H1), and spending bytes on witnesses of hard cases raised the share of decided hard cases over

Table 6: Pooled levels of every design at twelve messages per day (same columns as Table 5).

Design	Record	Hard, configured	Hard, unanticipated	All, configured
Independent loss, $Q = 12$				
Record only	0.972	0.759	0.641	0.950
3b rule	0.941	0.971	0.828	0.981
Exact allocator	0.956	0.969	0.742	0.994
Tolerance x 0.5	0.916	0.760	0.733	0.927
Send-on-delta, coded	0.864	0.610	0.570	0.857
Send-on-delta, once	0.249	0.405	0.380	0.509
Decision flags	0.969	1.000	0.275	1.000
Random-day control	0.946	0.887	0.766	0.972
Hybrid 1	0.969	1.000	0.692	1.000
Hybrid 2	0.930	1.000	0.828	1.000
Gilbert loss, $Q = 12$				
Record only	0.954	0.721	0.610	0.926
3b rule	0.917	0.955	0.813	0.959
Exact allocator	0.937	0.937	0.718	0.974
Tolerance x 0.5	0.882	0.699	0.680	0.883
Send-on-delta, coded	0.828	0.547	0.518	0.807
Send-on-delta, once	0.408	0.384	0.369	0.529
Decision flags	0.949	0.997	0.274	0.996
Random-day control	0.922	0.857	0.741	0.949
Hybrid 1	0.949	0.997	0.669	0.997
Hybrid 2	0.905	0.997	0.815	0.997

the record alone by about 0.3 at six messages per day (H2), more than the same layering on random days (H3), at a cost of 0.07 to 0.08 in the share of days whose whole record was recovered (H4). The witnesses also decided thresholds that nobody configured, which the flags decided in only 0.275 of hard cases by monotone implication (H5), and neither send-on-delta nor a tightened tolerance reached the same combination of record recovery and decisions (H6). The study does not claim that witnesses are the best way to decide thresholds that are fixed when the node is configured: flags were better there in development and in confirmation (H7).

The practical reading is a choice between three designs, depending on how fixed the thresholds are:

- If the thresholds are known and stable, decision flags decide nearly every hard

Table 7: Levels at six messages per day by region (hard cases decided at unanticipated thresholds, then record recovered, separated by a bar; descriptive, four to nine sites per region).

Region	Record only	3b rule	Decision flags	Hybrid 1
Independent loss, $Q = 6$ (hard cases decided at unanticipated thresholds | record recovered)				
Gulf and lower Mississippi (4 sites)	0.436 | 0.654	0.849 | 0.488	0.368 | 0.635	0.578 | 0.635
Mid-Atlantic and Appalachia (9 sites)	0.656 | 0.945	0.848 | 0.883	0.255 | 0.934	0.708 | 0.934
Ohio valley and Great Lakes (9 sites)	0.614 | 0.887	0.810 | 0.803	0.268 | 0.867	0.679 | 0.867
Pacific coast (9 sites)	0.527 | 0.935	0.738 | 0.865	0.211 | 0.924	0.591 | 0.924
Gilbert loss, $Q = 6$ (hard cases decided at unanticipated thresholds | record recovered)				
Gulf and lower Mississippi (4 sites)	0.392 | 0.606	0.772 | 0.456	0.354 | 0.590	0.540 | 0.590
Mid-Atlantic and Appalachia (9 sites)	0.539 | 0.864	0.735 | 0.813	0.246 | 0.852	0.612 | 0.852
Ohio valley and Great Lakes (9 sites)	0.514 | 0.810	0.708 | 0.737	0.257 | 0.792	0.594 | 0.792
Pacific coast (9 sites)	0.434 | 0.859	0.627 | 0.800	0.202 | 0.848	0.514 | 0.848

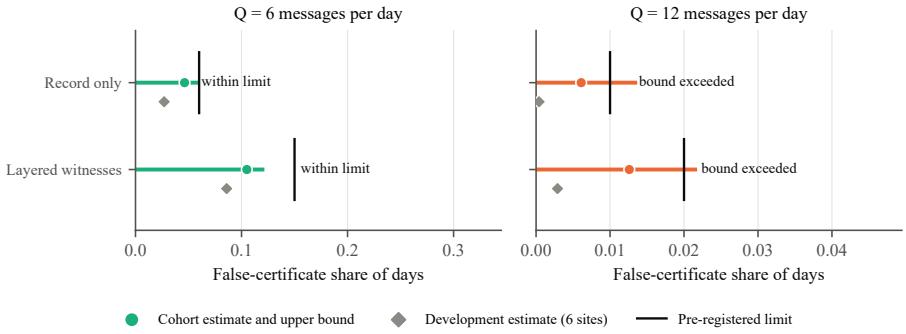


Figure 5: Share of days with a false certificate after one CRC-valid forgery, with the full receiver. Dots are cohort estimates with their upper bounds; diamonds are the development estimates from six sites; the vertical line is the pre-registered limit. The bounds at twelve messages per day exceeded their limits.

case (0.999 at independent loss) and recover the record almost as often as the record alone (0.877 against 0.892), but decide only 0.275 of hard cases at thresholds that were not configured, because they do not use the recovered record (the record alone decides 0.546).

- If thresholds may change, or if numerical extrema are wanted, hybrid 1 keeps the flag performance, adds witnesses in spare slots at no cost in record recovery or configured decisions, and raised unanticipated hard-case decisions from 0.275 to 0.629 (independent loss, descriptive), which is 0.083 above the record

alone (0.097 under Gilbert loss); no design of flags plus a record certificate without witnesses was run; therefore, the share due to the hosted witnesses alone was not measured. About 78% of development days had at least one spare slot and 21% had four (computed at the 4.75 capacity of the flags); therefore, the benefit depends on the room that the record leaves.

- If the unanticipated thresholds matter most, the targeted layered design (3b rule or hybrid 2) reached 0.806, at a cost in record recovery.

This is a claim about use, not a measured benefit in the field: the unanticipated threshold sets were chosen by the analyst, and whether thresholds change in practice was not studied.

6.2. The failed hypothesis

H8 failed at twelve messages per day, for record only and for layered witnesses, because the upper bounds of the conditional false-certificate share exceeded their limits (0.0134 against 0.01, and 0.0215 against 0.02). The development estimates, from six sites, were lower than the cohort values by a factor of 1.2 to 15. Two lessons follow:

- Corroboration: the corroboration rule compares surplus messages; therefore, its value rises with Q, but it cannot act when no more messages arrive than are needed, when fewer arrive and only systematic units are used, or when the record exceeds the quota and carries no parity. At $Q = 12$ the last two cases carried most of the false-certificate rate.
- Sample: a sample of six sites held almost none of the days with fewer messages than needed, and it understated a conditional rate that depends on the mix of days.

The consequence for use is that the CRC-16 and the free receiver rules are not shown to be sufficient at the stated limits. A longer check (CRC-24) or a message authentication code lowers the probability that a corrupted message is accepted, and it does not change the conditional share of H8. The conditional share is a property of days on which no redundancy exists to compare. A check that spans the whole record can verify a day only when every unit was decoded, which helps the days with exactly the needed number of messages; it cannot verify units that never arrived, and uncoded oversize days have no parity. For those days only more check bits per message lower the rate, and a metric that conditions on passing the check does not measure that gain; an unconditional rate at a stated bit-error rate does. None of these measures was tested.

6.3. Limits

- Simulation only: the evidence is trace-driven replay of USGS records through modelled loss. No satellite hardware, no measured channel error statistics, and no microcontroller measurements were used.
- Scope: 31 rivers of the United States in one year, with only four Gulf sites; nothing is claimed for other regions, years, or deployments.
- Assumption R: it was off for about 15 of 38 development sites, and it was inferred from the same record that it protects. H1 is therefore a test of the code, conditional on R.
- Flags capacity: flags were modelled at 4.75 record values per message although the frame allows 4. This favours the flags, and it favours hybrid 1 more than the flags on the unanticipated-threshold contrast, because it enlarges the spare room for hosted witnesses; the size of the effect was not measured.
- Slot model: the design comparisons used an abstract slot model and slotted witnesses; hosted witnesses, the conductance pair codes, the flags, and decoding from a subset of messages had no byte-level test in the confirmation.
- Construction: the contrasts of H7 and the configured and record parts of H9 hold by construction; therefore, the count of supported hypotheses overstates the evidence; the informative ones are H2 to H5, the unanticipated part of H9, and H8.
- Integrity model: forgeries changed body bytes only; false means readings certified outside tolerance or witness codes that differ from the truth, not wrong decisions.
- Run record: three environment variables of the method code (the 7DADM band, the unknown-history rule, and a day limit) and the library versions were not recorded, and the freeze record was committed after the run.
- Analyst choices: the alternative threshold sets, the hard-case bands, the 7DADM band, and the unknown-history rule were chosen by the analyst, the last two on development data after a first run fell short.
- Truth: truth statements refer to the loader's record, which includes readings bridged over gaps of at most two readings, and not to the river.
- Witness decisions are a lower bound on what witnesses could decide, because the box rule is not exact with witnesses (Proposition 2).
- Integrity: the forgery model uses random content conditional on a CRC pass; an adversary needs authentication, which is out of scope.
- Review: the code, the proofs, and the plan were read by one separate AI session, which is an in-project reading and not an outside review, and by the project team; no outside reviewer has read them. The tests were checked by a mutation

check of 18 hand-chosen faults, which found six test gaps that were then closed.
- Priority: the novelty statement of Section 2 rests on a general web search and on sources read in part; it is not a systematic review.

7. Conclusion

This study changed the question for river water-quality telemetry over a one-way satellite link from how to deliver the whole record to what the server can soundly decide from what arrives. A certificate language turns each piece of received information into an interval and answers threshold, extremum, conductance-change, and rolling-mean queries with exceeded, not exceeded, or undetermined; the answers are sound when the intervals are valid, exact only without witnesses, and limited by the tolerance for readings that lie near a threshold.

In a pre-registered confirmation on 31 sealed rivers, eight of nine hypotheses held, although some of the eight could not fail by design (H1 through assumption R, H7, and most of H9) and the informative ones are H2 to H5, the unanticipated part of H9, and H8. No certified decision was wrong. Targeted witnesses raised the share of decided hard cases over a record alone by 0.30 to 0.32 at six messages per day and 30% loss, and decided thresholds that nobody configured. Decision flags remained better for thresholds fixed at configuration time, but they decide from the flags alone; a hybrid that hosts witnesses in the spare slots of the flag design cost nothing on configured thresholds and decided 0.08 to 0.10 more hard cases than the record alone, and 0.29 to 0.36 more than the flags, on unanticipated thresholds. The hypothesis on false certificates after a forged message that passes the 16-bit check failed at twelve messages per day, and the failure points to days on which no message can be compared, which a longer check does not remove.

The claim of this study is generality and soundness, not the best decisions for known thresholds, and the evidence is trace-driven simulation. Three steps would move it toward practice: a measurement of real channel loss and error statistics, a receiver design whose integrity is measured as an unconditional rate at a stated bit-error rate, with a record-level check and a longer per-message check, tested on a new cohort, and an outside review of the code and proofs. The software, the frozen plan, and every result file are in the project repository.

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